PATTERN DETECTOR

Black-Box Verification Specification

UCIe 3.0 | Sideband Receiver (SB_RX)

Document Version	1.0
Specification Basis	UCIe Revision 3.0, Version 1.0 (August 5, 2025)
Audience	Design Verification / UVM Engineers

Purpose

This document defines the externally observable behavior of the SB_RX Pattern Detector and provides a verification-oriented interface contract, stimulus rules, packet construction reference, directed test plan, and coverage checklist. The detector is treated as a black box: the verifier drives its documented inputs and checks its outputs without depending on internal counters or implementation structure.

At a glance

Item	Black-box contract
Primary functions	Detect a fixed 64-bit training pattern; detect the SBINIT Out of Reset sideband message.
Data interface	64-bit parallel input plus data_valid.
Clock / reset	sb_clk; sb_sys_rst is active-low asynchronous reset.
Output events	pattern_detected and sb_o_detected.
State windows	pattern_detected: RESET and SBINIT. sb_o_detected: SBINIT only.

Important usage rule The detector does not perform the Controller state transition. It reports detection events; the SB_RX Controller consumes synchronized / qualified versions and owns the link-training state progression.

1. Scope and SB_RX integration

The Pattern Detector belongs to the startup branch of SB_RX. Incoming sideband serial data is first assembled by the deserializer into 64-bit words. A 1-to-2 demultiplexer selects either the startup-detection branch or the normal-message receive branch.

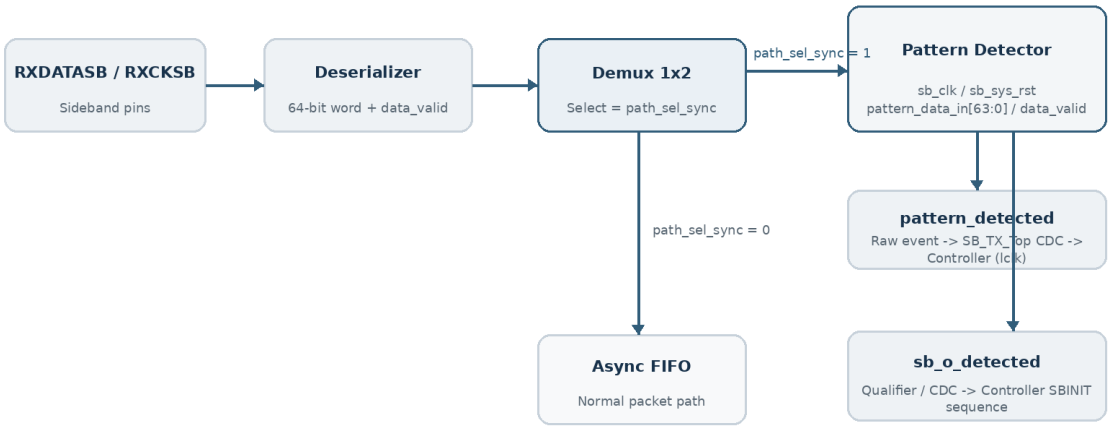


Figure 1. Pattern Detector placement in the SB_RX receive flow (logical view).

Selector condition	Selected path	Expected detector activity
path_sel_sync = 1	Demux output 1 -> Pattern Detector	Detector receives pattern_data_in and data_valid.
path_sel_sync = 0	Demux output 0 -> Asynchronous FIFO	Pattern Detector is not selected by the top-level receive path.

Boundary distinction path_sel_sync is a top-level path-selection signal, not a listed Pattern Detector input. A standalone detector test cannot directly test path selection unless a wrapper exposes it; verify that behavior in an SB_RX integration test.

2. DUT interface

Port / parameter	Direction	Width	Description / verifier usage
DATA_WIDTH	Parameter	64	Width of pattern_data_in. The supplied block description uses DATA_WIDTH = 64.
sb_clk	Input	1	Sideband receive clock. Detection logic is evaluated in this clock domain.
sb_sys_rst	Input	1	Active-low asynchronous reset from the interface.
pattern_data_in	Input	DATA_WIDTH	Parallel data from Demux1x2. Drive stable around the active sb_clk sampling edge when data_valid = 1.
data_valid	Input	1	Deserializer valid indicator. Only valid samples participate; invalid data must not create a detection event.
pattern_detected	Output	1	Detection indication for the fixed training pattern.
sb_o_detected	Output	1	Detection indication for the SBINIT Out of Reset message.

3. Functional behavior

3.1 Fixed training-pattern detection: pattern_detected

Property	Expected behavior
Recognized word	64'h5555_5555_5555_5555
Qualification	The block description states that two consecutive valid 64-bit inputs must match the fixed pattern.
Assertion point	pattern_detected is asserted on the second qualifying occurrence, not on the first.
State enable	Enabled only during RESET and SBINIT.
Output meaning	An event/flag indicating pattern recognition; it is not the data payload.

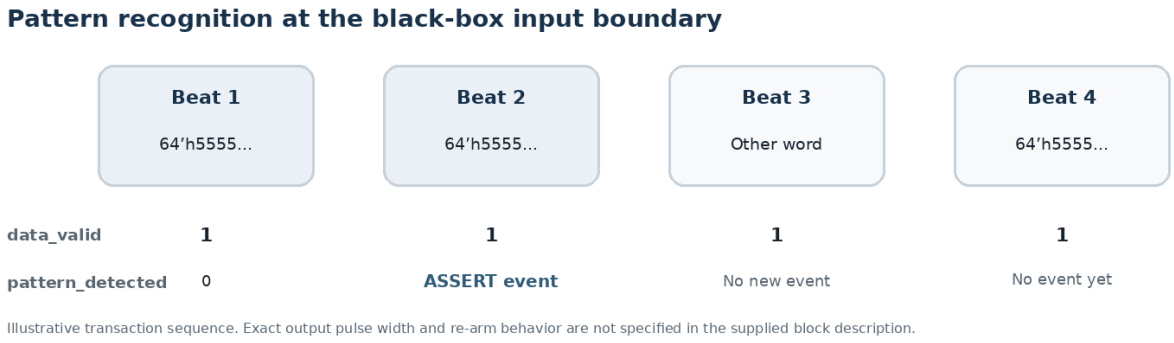


Figure 2. Representative two-match recognition sequence. The post-event pulse/hold behavior is not specified by the supplied block description.

Consecutive-match interpretation A non-matching valid word should break the consecutive-match run under the ordinary meaning of "two consecutive valid inputs". Whether a cycle with data_valid = 0 breaks the run is not stated in the supplied description; track it as a specification question.

3.2 SBINIT Out of Reset message detection: sb_o_detected

- The output indicates recognition of a correctly received {SBINIT Out of Reset} sideband packet.
- This detection is active only while the LTSSM is in SBINIT.
- The packet is a message without data. Its protocol encoding is described in UCIe 3.0 Table 7-9 and Figure 7-3.
- The project-level RTL analysis reports checks on the opcode, MsgCode 8'h91, selected fields, and parity. It also notes that full MsgSubcode and Result validation may be missing in the reviewed RTL snapshot. This is an implementation observation, not a relaxation of the protocol packet definition.

3.3 State enable matrix

LTSSM state	pattern_detected	sb_o_detected	Verification intent
RESET	Enabled	Disabled	Training-pattern detection only.
SBINIT	Enabled	Enabled	Training-pattern detection and SBINIT Out of Reset packet recognition.
All other states	Disabled	Disabled	No detection event should be generated outside the documented enable window.

State visibility No state port appears in the provided Pattern Detector interface table. Exercise the state matrix through the SB_RX top-level/controller or through a verification wrapper that exposes the actual internal enable. Do not invent a new DUT port.

4. SBINIT Out of Reset packet definition

The {SBINIT Out of Reset} message is a UCIe message-without-data packet. The following signature is the protocol-level reference for stimulus generation and scoreboard classification.

Field	Width	Required / defined value	Meaning
Opcode	5	10010b	Message without Data packet type.
srcid[2:0]	3	010b	Physical Layer source.
dstid[2:0]	3	110b	Remote-die-terminated Physical Layer message: dstid[2] = 1 and dstid[1:0] = 10b.
MsgCode	8	91h	SBINIT Out of Reset message code.
MsgSubcode	8	00h	SBINIT Out of Reset message subcode.
MsgInfo[15:4]	12	Reserved	Reserved portion of MsgInfo.
MsgInfo[3:0]	4	Result[3:0]	Sideband data/clock detection result mask.
DP	1	0	No data payload; DP is 0 when there is no data payload.
CP	1	Even parity	Control parity over header bits excluding DP.

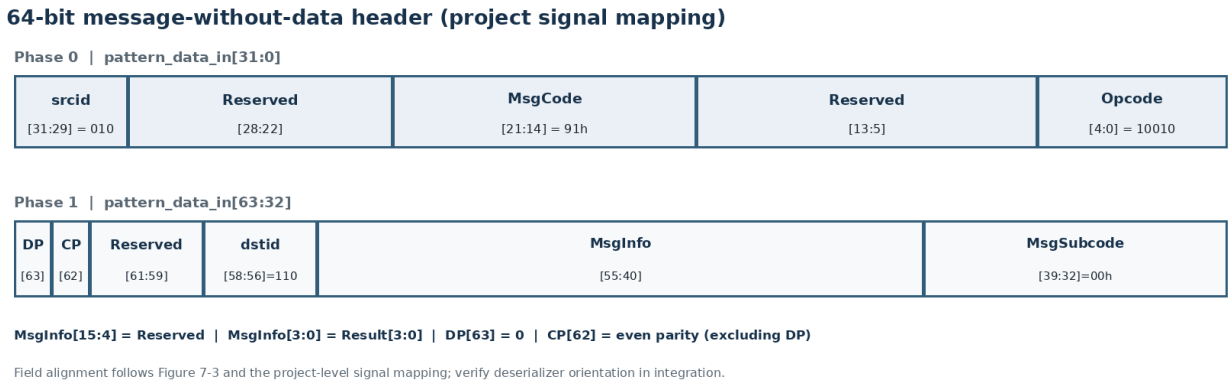


Figure 3. Packet field mapping for the project pattern_data_in[63:0] signal. Phase 0 occupies [31:0]; Phase 1 occupies [63:32].

4.1 Result[3:0] interpretation

Result bit	Receiver combination represented	1 means
Result[0]	DATASB sampled with CKSB	Pattern detected.
Result[1]	DATASB sampled with CKSBRD	Pattern detected.
Result[2]	DATASBRD sampled with CKSB	Pattern detected.
Result[3]	DATASBRD sampled with CKSBRD	Pattern detected.

Result policy to confirm The UCIe initialization text uses Result[3:0] to select a functional sideband combination and states that no detected combination means the sideband is not functional. Confirm whether this detector must reject Result = 0000b, or whether it only recognizes the message encoding and leaves result handling to the Controller. For stimulus, set DP[63] = 0, calculate CP[62] as even parity over the header excluding DP, apply the agreed reserved-bit policy, and verify bit orientation at integration.

5. Clock, reset, timing, and system integration

Topic	Known contract	Verification guidance
Clock domain	sb_clk drives the detector logic.	Drive and sample at sb_clk. Keep input stable around the sampling edge.
Reset	sb_sys_rst is active-low and asynchronous.	Assert reset between partial matches and verify the detector does not retain stale progress. Confirm exact reset values and release behavior in RTL.
Output domain	pattern_detected and sb_o_detected originate in the detector domain.	Do not compare the raw detector output against lclk-domain controller timing.
CDC to Controller	The SB_RX flow uses synchronizers to deliver detector events into lclk.	Verify CDC latency/handshake separately at integration level; do not attribute synchronizer latency to the detector itself.
Pattern path enable	Top-level path selection routes data to the detector when path_sel_sync = 1.	At integration level, confirm normal FIFO path is selected when path_sel_sync = 0.
Message event qualification	Top-level flow notes show sb_o_detected qualified by sb_o_detected_delay before Controller consumption.	Verify the actual qualification and synchronizer ordering against RTL wiring.

5.1 Timing points not specified by the supplied description

- Exact cycle latency from the second matching input word to pattern_detected.
- Whether each output is a one-cycle pulse, a sticky flag, or held until state transition / reset.
- How quickly the detector re-arms after a successful detection and whether repeated detections are permitted while remaining in the same state.
- Whether data_valid = 0 pauses or clears a partial pattern-match sequence.
- Whether reset assertion immediately forces both outputs low and clears every partial-match / message-candidate state.

Checker implementation Start with transaction-level checking: count only valid 64-bit samples and check that the detection event cannot occur before the second qualifying pattern word. After confirming RTL latency and pulse semantics, tighten the checker to exact cycle-level assertions.

5.2 Protocol-level versus detector-level scope

UCIe 3.0 describes the physical SBINIT pattern in UI terms: a 64-UI alternating clock pattern followed by 32 UI low, with successful detection defined by 128 UI of the clock pattern. The Pattern Detector block description supplied for this project instead specifies a 64-bit deserialized word comparison performed on two consecutive valid inputs. These are different abstraction levels.

- Detector unit test: verify the 64-bit word-level contract at pattern_data_in/data_valid.
- Deserializer / PHY integration test: verify that the serial UI pattern is sampled, assembled, and presented to this block with the intended bit ordering and valid cadence.
- Do not infer that two matching 64-bit words alone prove compliance with the full serial UI timing requirement.

6. Directed verification plan

The tests below are black-box scenarios. They intentionally target observable requirements and known ambiguity points; they do not require access to internal counters.

ID	Test	Priority	Stimulus	Expected result
PD-PAT-01	Reset baseline	Very High	Assert sb_sys_rst low. Hold data_valid low and vary pattern_data_in.	Outputs remain inactive; no event is reported during reset. Exact reset values to be confirmed.
PD-PAT-02	Single matching word	Very High	In RESET, drive one valid 64'h5555... word only.	pattern_detected must not assert on the first matching occurrence.

ID	Test	Priority	Stimulus	Expected result
PD-PAT-03	Two consecutive matches	Very High	In RESET, drive two consecutive valid exact pattern words.	pattern_detected asserts on the second qualifying occurrence.
PD-PAT-04	Non-match breaks run	Very High	Drive match, then a different valid word, then one match.	No detection; a new two-match run is required.
PD-PAT-05	Invalid beat isolation	High	Drive arbitrary data while data_valid = 0, including the pattern value.	Invalid samples must not create a detection event. Bubble effect on an existing partial run is an open item.
PD-PAT-06	One-bit near match	High	Drive exact pattern with each bit position flipped, one case per bit.	A near-match must not be treated as the exact fixed pattern.
PD-PAT-07	Reset clears partial run	High	Drive first match, assert reset, release reset, then drive one match.	Pre-reset partial progress must not combine with post-reset input.
PD-PAT-08	State gating	Very High	Repeat pattern sequence in RESET, SBINIT, and a non-enabled state.	Event allowed only in RESET/SBINIT.

Priority interpretation Very High = required for basic functional sign-off. High = important corner / robustness coverage. These priorities are verification planning recommendations, not UCIe-defined priorities.

6. Directed verification plan (continued)

ID	Test	Priority	Stimulus	Expected result
PD-SBO-01	Valid SBINIT Out of Reset packet	Very High	In SBINIT, drive a valid message-without-data packet with opcode 10010b, srcid 010b, dstid 110b, MsgCode 91h, MsgSubcode 00h, legal Result, DP=0, and correct CP.	sb_o_detected asserts after the valid packet is accepted.
PD-SBO-02	Wrong message code	Very High	Keep the rest of the header valid; change MsgCode from 91h.	sb_o_detected does not assert.
PD-SBO-03	Wrong opcode / packet type	Very High	Use an opcode other than 10010b with otherwise similar fields.	sb_o_detected does not assert.
PD-SBO-04	Wrong source / destination	High	Change srcid from 010b and/or dstid from 110b.	No false SBINIT Out of Reset detection.
PD-SBO-05	Wrong subcode	High	Use MsgSubcode other than 00h.	No false detection if full message signature is required; compare against reviewed RTL and log any gap.
PD-SBO-06	Parity corruption	Very High	Flip CP while holding all other fields constant. Also test DP=1 for this no-data packet.	Bad CP must be rejected; DP should be 0 for a no-data packet.
PD-SBO-07	Result mask sweep	High	Run Result = 0001, 0010, 0100, 1000, multi-hot, and 0000.	Check detection / rejection according to the agreed Result policy; record selected data/clock combination.
PD-SBO-08	State gating	Very High	Send the correct packet in RESET, SBINIT, and a non-enabled state.	sb_o_detected is permitted only in SBINIT.
PD-SBO-09	data_valid qualification	High	Present a correct packet while data_valid = 0, then repeat with data_valid = 1.	No detection for invalid presentation; detection only for valid packet.
PD-SBO-10	Confusing SBINIT messages	High	Send SBINIT done req (95h/01h), SBINIT done resp (9Ah/01h), and other training messages.	These messages must not be misclassified as SBINIT Out of Reset.

6.1 Integration-level tests

ID	Scenario	Expected result
PD-INT-01	path_sel_sync = 1 during startup	Valid deserializer words reach the detector branch; normal FIFO write request is gated off.
PD-INT-02	path_sel_sync = 0 during normal reception	Detector path is not selected; valid words are routed to the Async FIFO.
PD-INT-03	pattern_detected CDC	Raw detector event is observed in sb_clk domain; synchronized event reaches Controller in lclk domain after synchronizer latency.
PD-INT-04	sb_o_detected qualification / CDC	Top-level qualification and synchronization deliver only the intended SBINIT message event to the Controller.

7. Scoreboard and coverage guidance

7.1 Recommended black-box reference model

- Maintain a transaction counter for exact-pattern valid samples. The model must not rely on an internal DUT match counter.
- Under the strict “two consecutive valid inputs” interpretation, clear the match run on a non-matching valid word. Implement bubble handling only after the design owner confirms whether an invalid cycle pauses or clears the run.
- For sb_o_detected, decode the documented packet fields from the 64-bit word and check the full message signature, DP, CP, and Result policy agreed for this design.
- Keep state gating, top-level path selection, and CDC checking in an integration scoreboard/monitor when those signals are outside the detector boundary.

```
PATTERN = 64'h5555_5555_5555_5555;

if (!sb_sys_rst) begin
 pattern_run = 0;
 // expected outputs return to their reset state (confirm exact value)
end
else if (data_valid) begin
 if (pattern_data_in == PATTERN) pattern_run++;
 else                            pattern_run = 0;
 // pattern_detected event is expected on qualifying match #2
end
```

Reference model status The pseudocode is a verification interpretation of the supplied two-consecutive-valid-word description. It is not a substitute for confirming valid-bubble semantics, output latency, output pulse width, and state-enable connectivity.

7.2 Functional coverage model

Coverage group	Suggested bins / crosses
State	RESET, SBINIT, other; cross with each detection function.
Pattern input	Exact pattern, one-bit near-match, multi-bit mismatch, random word.
Valid cadence	Back-to-back valid, one-cycle bubble, multi-cycle bubble, invalid-only data.
Match history	No prior match, one prior match, interrupted match, reset after first match.
SBINIT message fields	Opcode match/mismatch; srcid match/mismatch; dstid match/mismatch; MsgCode match/mismatch; MsgSubcode match/mismatch.
Integrity fields	CP good/bad; DP=0/1; reserved-bit policy cases.
Result mask	0001, 0010, 0100, 1000, multi-hot, 0000.

Coverage group	Suggested bins / crosses
Output behavior	No event, first-match suppression, successful event, repeated candidate, post-reset behavior.

7.3 Sampling recommendations

- Use a clocking block or equivalent race-free sampling discipline on sb_clk.
- Drive data_valid and pattern_data_in before the sampling edge and hold them stable through the edge.
- Record input transactions and output transitions with cycle timestamps. Keep the output latency window configurable until the RTL contract is finalized.
- For integration, monitor both raw detector outputs and their synchronized Controller-side versions. Do not conflate the two timing points.

8. RTL observations and specification questions

The supplied SB_RX flow guide contains several observations from the reviewed RTL snapshot. These should be treated as verification targets and design-review findings, not as normative UCIe requirements.

Item	Observation / open question	Recommended action
Consecutive-match counter	The flow guide reports that the match counter is not cleared on every non-matching valid word. This may allow non-consecutive matches to accumulate.	Run PD-PAT-04. Confirm expected consecutive semantics and raise a design issue if RTL violates it.
Full physical pattern	The RTL analysis notes that the word comparator does not explicitly verify the complete low-UI interval described by the physical SBINIT pattern.	Keep UI-level validation at PHY/deserializer integration. Do not claim word-level detection proves serial-level compliance.
Message field coverage	The flow guide reports that selected fields and parity are checked, while full MsgSubcode and Result checking may be absent.	Run PD-SBO-05 and PD-SBO-07. Compare the actual checker against the agreed full packet signature.
LTSSM state visibility	No LTSSM state input appears in the provided detector port table, even though enable conditions are state-dependent.	Confirm whether state gating is internal or achieved through upstream valid/branch gating.
Output timing	Pulse width, sticky behavior, and exact output latency are not defined in the supplied description.	Agree a timing contract with the RTL owner and then add cycle-accurate assertions.
Valid bubbles	The effect of data_valid=0 between two matching words is not defined.	Add a directed bubble test and freeze the expected result after design clarification.
Result=0000	UCIe describes no selected sideband combination as non-functional, but the detector acceptance/rejection rule is not explicit.	Confirm whether sb_o_detected is purely a message recognizer or also validates functional Result content.

9. Sign-off checklist

- ☐ Interface widths, clock, and active-low asynchronous reset have been checked against the RTL.
- ☐ One matching pattern word does not trigger pattern_detected; the second qualifying word does.
- ☐ A non-matching valid word breaks the pattern run, or any deviation is explicitly approved.
- ☐ Invalid input cycles cannot create a false detection event; bubble behavior is documented.
- ☐ Pattern detection is enabled only in RESET/SBINIT; SBINIT Out of Reset detection is enabled only in SBINIT.
- ☐ A valid SBINIT Out of Reset packet is recognized using the approved header fields and parity rules.
- ☐ Wrong opcode, source, destination, MsgCode, MsgSubcode, and bad parity are covered.
- ☐ Result mask interpretation is documented and covered.
- ☐ Reset clears partial detection history and outputs according to the agreed reset contract.
- ☐ Top-level path selection and CDC/qualification are verified separately from the detector unit test.

10. Source references and interpretation notes

Source	Used for
Pattern Detector block description supplied as screenshot: Section 3.3.2.2, Figure 3-34, Tables 3-55 and 3-56.	Module purpose; DATA_WIDTH=64; port names/directions; active-low asynchronous reset; fixed pattern; two-consecutive-valid-input behavior; state enable conditions.
UCIe Specification Rev. 3.0, Version 1.0, August 5, 2025.	Section 4.5.3.2, Sideband Initialization (SBINIT), pp. 133-135: UI-level pattern and initialization sequence; Result[3:0] mapping and post-detection message flow.
UCIe Specification Rev. 3.0, Version 1.0.	Section 7.1.2.2, Figure 7-3, p. 275: message-without-data packet format; Table 7-4, p. 271: srcid/dstid encodings; Table 7-5, p. 272: CP/DP parity definitions; Table 7-9, p. 278: SBINIT Out of Reset message encoding.
SB_RX_Main_Flow_Guide(2).html supplied by user.	Top-level branch selection, detector output consumers, clock-domain context, and RTL observations on pattern counter and message-field checking.
SB_RX_TOP_Analysis(1).html supplied by user.	SB_RX training flow context, controller-side signal routing, and integration verification considerations.

Interpretation convention

- Protocol / block-description statements are treated as the intended verification contract.
- RTL observations describe the reviewed source snapshot and should be converted into pass/fail requirements only after design-owner confirmation.
- Open items are intentionally visible so that the verifier does not invent cycle timing, state wiring, or packet acceptance rules that are not present in the supplied material.

Specification provenance The UCIe specification is referenced by revision, section, table, and page. This document paraphrases the relevant requirements and provides project-specific verification guidance; it is not a replacement for the UCIe specification.